

Load Interconnection Request System Impact Study Report

For Queue Project Q11-0073

Jones Chapel 500 kV
November 26, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0073. The Interconnection Customer proposes a 672.7 MW large load (data center) facility interconnecting at the Jones Chapel 500 kV substation in the Indiana area. The request was received on September 29, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 64 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$91,246,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Swan Vly Substation – Twin County Ranch Substation 230kV	Trego – Swan Vly 500kV	125.1	439	8.6
Monfort – Wabash 230kV line	Montpelier Electric Generating Station – Waynedale 230kV	112.2	818	11.1
Montpelier Electric Generating Sta – Waynedale 230kV L	Wabash 500/230kV Autotransformer	111.3	809	11.7
Mcneil Substation – Breed 230kV	Richland – Dekalb 500kV line	108.0	513	27.3
Breed – Sidney A Murray Jr Hydroelectric 230kV	Richland – Dekalb 500kV	107.1	518	27.3
State Street – Waynedale 230kV	Richland – Dekalb 500kV	104.7	692	17.3
Monfort – Waynedale 230kV	Montpelier Electric Generating Sta – Wynedale 230kV	102.0	820	10.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Swan Vly Substation – Twin County Ranch Substation 230kV	Rebuild line with higher-capacity conductor	\$28,258,000	60% (\$16,992,000)
Monfort – Wabash 230kV line	Rebuild line with higher-capacity conductor	\$34,554,000	47% (\$16,128,000)
Waynedale – Montpelier Electric Generating Sta 230kV line	Rebuild line with higher-capacity conductor	\$71,471,000	47% (\$33,356,000)
Mcneil Substation – Breed 230kV	Reconductor line	\$8,241,000	55% (\$4,504,000)
Breed – Sidney A Murray Jr Hydroelectric 230kV	Reconductor line	\$18,410,000	55% (\$10,062,000)
Waynedale – State St 230kV	Reconductor line	\$4,458,000	52% (\$2,296,000)
Monfort – Waynedale 230kV	Reconductor line	\$16,863,000	47% (\$7,908,000)
	Total allocated to project		\$91,246,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.